

Customer Subscription Retention Prediction Using DNA Computing & CNN

Project Phase-2 Report

Submitted in partial fulfillment of the requirements for the award of degree of

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in

Computer Science and Engineering (Data Science)

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BONAFIDE CERTIFICATE

This is to certify that the Project Phase-1 report on the topic “**Customer Subscription Retention Prediction Using DNA Computing & CNN**” by **Anurag K P (Reg No:SCM22CD017), Fazil Firose Ibrahim(Reg No:SCM22CD032), Krishnendu C (Reg No:SCM22CD042), Parvathi Prakash Menon (Reg No: SCM22CD053)** , to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science & Engineering (Data Science) is a Bonafide record of work carried out under our guidance and supervision during the academic year 2025-26.

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DECLARATION

We undersigned hereby declare that the project phase-II report Customer Subscription Retention Prediction Using DNA Computing & CNN, submitted for complete fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering(Data Science) of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Dr. Sonal Ayyappan. This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately and accurately cited and referenced the sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University

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ABSTRACT

Customer subscription retention prediction plays a vital role in sustaining digital businesses, as retaining existing customers is significantly more cost-effective than acquiring new ones. However, accurately identifying customers likely to discontinue subscriptions remains a challenge due to complex, nonlinear behavioral and transactional patterns. This project proposes an approach that integrates DNA Computing with Convolutional Neural Networks (CNN) to enhance prediction accuracy and interpretability. DNA computing is utilized to encode customer features and behaviors into DNA-like sequences, optimizing data representation and feature selection. These sequences are transformed into matrices suitable for CNN processing, enabling the model to learn intricate dependencies in customer data. The system architecture includes a Flask-based backend, React.js frontend dashboard, and all containerized using Docker for efficient deployment. Experimental evaluation using benchmark churn datasets shows that the DNA–CNN model achieves superior predictive performance compared to traditional methods, allowing for early identification of at-risk customers and supporting data-driven, cost-effective retention strategies.

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Chapter 1

INTRODUCTION

In today's competitive digital landscape, customer subscription retention is pivotal for the sustained success of online service platforms, as retaining existing customers is considerably more cost-effective than acquiring new ones. Accurately predicting which customers are likely to discontinue their subscriptions is a significant challenge due to the complex, nonlinear dependencies inherent in behavioral and transactional data. Addressing these limitations, this project introduces a novel framework that combines DNA computing—a biologically inspired approach for data encoding and feature selection—with Convolutional Neural Networks (CNN), renowned for their capacity to model high-dimensional patterns. By integrating DNA-based data representation with deep learning, the system seeks to improve churn prediction accuracy, empower proactive retention strategies, and deliver scalable, data-driven insights for modern subscription businesses. The solution leverages a modular architecture including a Flask backend, React.js frontend dashboard, and Docker-based deployment to ensure robust real-time performance and seamless scalability.

1.1 Overview

Customer subscription retention has become a key factor determining the long-term success of digital businesses and online service platforms. Retaining existing customers is significantly more cost-efficient than acquiring new ones, yet accurately predicting which customers are at risk of discontinuing their subscriptions remains a complex challenge. Traditional machine

learning and statistical models often struggle to capture the nonlinear relationships present in customer behavior, transaction history, and service usage patterns. These limitations create the need for advanced computational techniques capable of extracting deeper insights from large and diverse data sources.

This project introduces a novel framework that combines DNA Computing with Convolutional Neural Networks (CNN) to improve customer retention prediction accuracy. DNA computing offers a biologically inspired approach to data encoding and feature optimization by representing customer attributes as DNA-like sequences, enhancing data diversity and interpretability. These encoded sequences are then processed by a CNN, which excels at identifying spatial and sequential relationships in high-dimensional data. By merging these two paradigms, the system aims to provide a more accurate, robust, and efficient prediction model.

The architecture integrates multiple components, including a Flask-based backend for model deployment and a React.js frontend dashboard for visualization of churn predictions. The entire solution is containerized using Docker, enabling seamless deployment and scalability. Overall, this approach empowers businesses to proactively identify at-risk customers, reduce churn, and design targeted retention strategies driven by advanced insights.

1.2 Problem Statement

This project aims to develop a computational model integrating DNA Computing and Convolutional Neural Networks to improve accuracy in predicting customer subscription retention. It addresses the challenge of capturing complex, nonlinear customer behavior from diverse data sources to enable timely, reliable churn prediction and effective retention.

1.3 Objective

The primary objective of this project is to develop a DNA Computing and CNN model that enhances the prediction accuracy of customer subscription retention. Specific goals include:

- Encoding customer data into DNA-like sequences for optimized feature representation.
- Training a CNN model capable of learning intricate behavioral and transactional patterns.
- Building a full-stack system integrating backend, frontend, and CNN for real-time churn prediction.
- Evaluating the model's performance against traditional methods to validate improvements in accuracy and efficiency.

Through this approach, the project aims to deliver a scalable, intelligent, and data-driven solution for proactive customer retention management in subscription-based industries.

Chapter 2

LITERATURE REVIEW

DNA computing and security have become major interdisciplinary fields combining molecular biology, data science, and artificial intelligence. Recent studies reveal DNA's capability for computation, data storage, cryptography, and threat detection. This survey highlights four significant works that explore computational design, information protection, and DNA-based security systems.

Liu et al. [1] proposed a DNA-based molecular framework integrating computing, storage, and communication. The study addressed the scaling limits of silicon technology and introduced DNA as a molecular substrate offering massive parallel computation, dense storage, and low energy usage. The authors implemented strand displacement reactions (SDR) to realize Boolean logic gates such as AND, OR, NOT, and XOR within DNA molecules. Their experiments showed efficient molecular circuits and reliable DNA data storage with error correction. The work also suggested bio-electronic systems combining molecular logic with digital processors for scalable, sustainable computation.

Khan et al. [2] developed DNAact-Ran, a digital DNA sequencing engine for ransomware detection using machine learning. The model represented software behavior as digital genomes, applying DNA sequencing operations to identify malicious traits. It incorporated Multi-Objective Grey Wolf Optimization (MOGWO) and Binary Cuckoo Search (BCS) algorithms for feature selection and dimensionality reduction, improving accuracy and efficiency. Tested on datasets of over 1,500 samples, the model outperformed traditional

classifiers such as Naïve Bayes and AdaBoost. This research introduced digital genotypes and phenotypes for ransomware family classification, merging bioinformatics and cybersecurity to enable proactive malware detection.

Mahjabin et al. [4] presented a survey on DNA-based cryptography and steganography, classifying them into natural DNA cryptography, pseudo-DNA cryptography, and DNA-based steganography. Natural DNA cryptography uses biochemical processes for encryption, while pseudo-DNA cryptography simulates these reactions computationally. DNA-based steganography conceals data within nucleotide sequences for secret transmission. The paper reviewed encoding and substitution techniques that enhance data confidentiality and resilience. Although DNA systems offer superior storage, parallelism, and durability, the authors noted challenges such as synthesis costs, decoding errors, and biological data vulnerability. They called for collaboration between biocomputing and cybersecurity to develop reliable and low-cost DNA security models.

Together, these studies highlight DNA's promise as a secure and powerful computational medium. They demonstrate that integrating molecular computation with artificial intelligence and cybersecurity can lead to scalable and resilient systems for next-generation information processing and protection.

Chapter 3

SYSTEM ARCHITECTURE AND METHODOLOGY

The proposed project employs a methodology that encodes customer behavioral and transactional features into DNA-like sequences for optimized data representation, which are then transformed into matrices suitable for input to a Convolutional Neural Network (CNN). The CNN model is trained to learn intricate dependencies and nonlinear patterns within the encoded data, enabling highly accurate prediction of customer subscription retention. The system is implemented as a full-stack solution, comprising a Flask-based backend for model deployment, a React.js frontend dashboard for visualization, and all containerized with Docker for efficient and seamless deployment. This integrated pipeline facilitates early detection of at-risk customers, supporting proactive retention strategies through advanced, interpretable analytics.

3.1 Methodology Steps:

- Collect and preprocess customer behavioral and transactional data to ensure quality and consistency.
- Encode the processed features into DNA-like sequences, optimizing data representation for subsequent modeling.
- Transform the encoded sequences into structured matrices suitable as input for Convolutional Neural Networks (CNN).

- Train a CNN model to identify complex nonlinear patterns and dependencies within the encoded customer data.
- Integrate the trained model into a full-stack system with a Flask backend and React.js dashboard for visualization.
- Evaluate model performance against traditional methods, using metrics such as accuracy and efficiency to validate the improved prediction and retention capabilities.

3.2 System Architecture

The proposed system architecture is modular and comprises three principal components: the frontend, the backend and the CNN model. The **frontend** includes interactive dashboard and visualization modules accessible to users, providing real-time churn prediction results, DNA pattern visualizations, and actionable retention suggestions. Communication between the frontend and backend is built with Flask, enabling secure prediction requests and the retrieval of results and DNA-based feature representations.

The **backend** is responsible for all core computational functionalities. Data preparation begins with customer data collection, followed by DNA encoding to transform behavioral and transactional features into specialized representations, and a 2D matrix transformation to format the data for deep learning inference. This processed input is sent to a Convolutional Neural Network (CNN), implemented using TensorFlow/Keras, which performs churn prediction based on learned patterns. Inference requests are managed through the API, and the resulting predictions are communicated back to the frontend, ensuring seamless, low-latency operation.

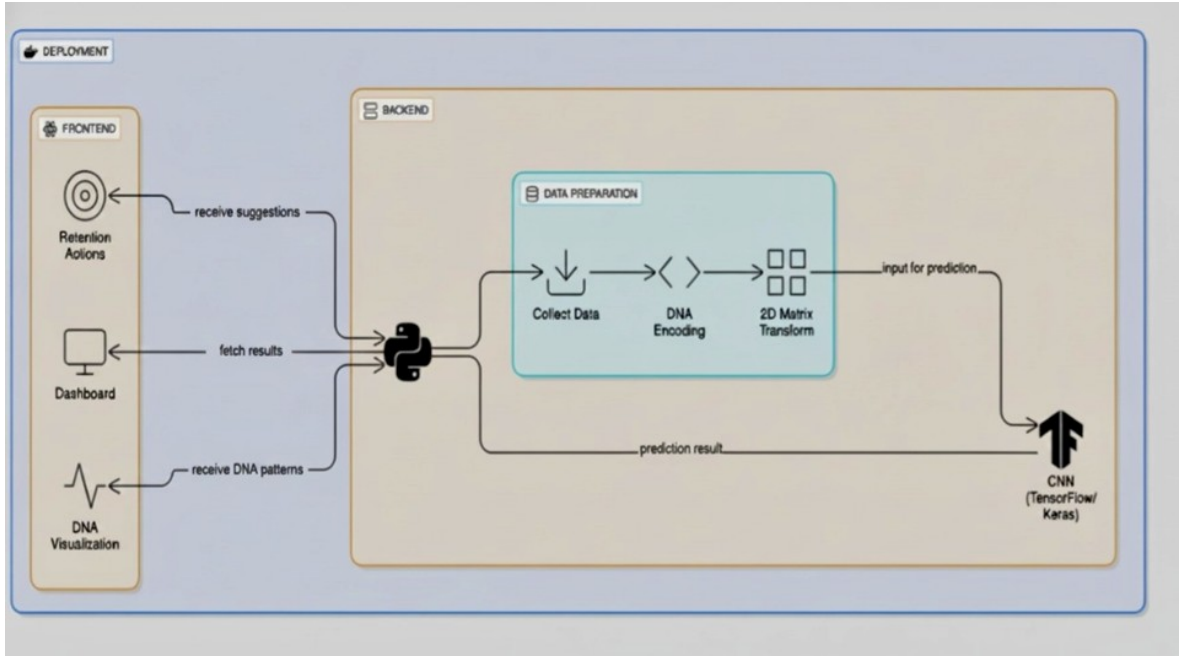


Figure 3.1: System Architecture

3.2.1 CNN Architecture

The convolutional neural network (CNN) architecture utilized for pattern prediction involves several sequential stages designed to effectively learn and classify complex data patterns. The process begins with the input dataset, which is typically preprocessed into a uniform format for analysis. Initial layers, known as convolutional layers, apply a series of filters with activation functions (such as ReLU) to the input data to extract low-level features like edges and textures, preserving the spatial relationships within the data. Pooling layers then downsample these feature maps, reducing dimensionality and highlighting the most important features for subsequent processing. The feature maps are subsequently flattened and passed through fully connected layers, which enable the network to learn complex, high-level representations and relations within the extracted features. Dropout regularization may be employed to improve model generalization. For the final classification, the learned feature representations are processed through the output layer to generate probabilistic predictions for each class. During training, an appropriate loss function like cross-entropy is used, and the network weights are

optimized via algorithms such as Adam to maximize predictive accuracy. This architecture ensures robust automatic feature learning, efficient dimensionality reduction, and effective pattern prediction for a wide range of data types.

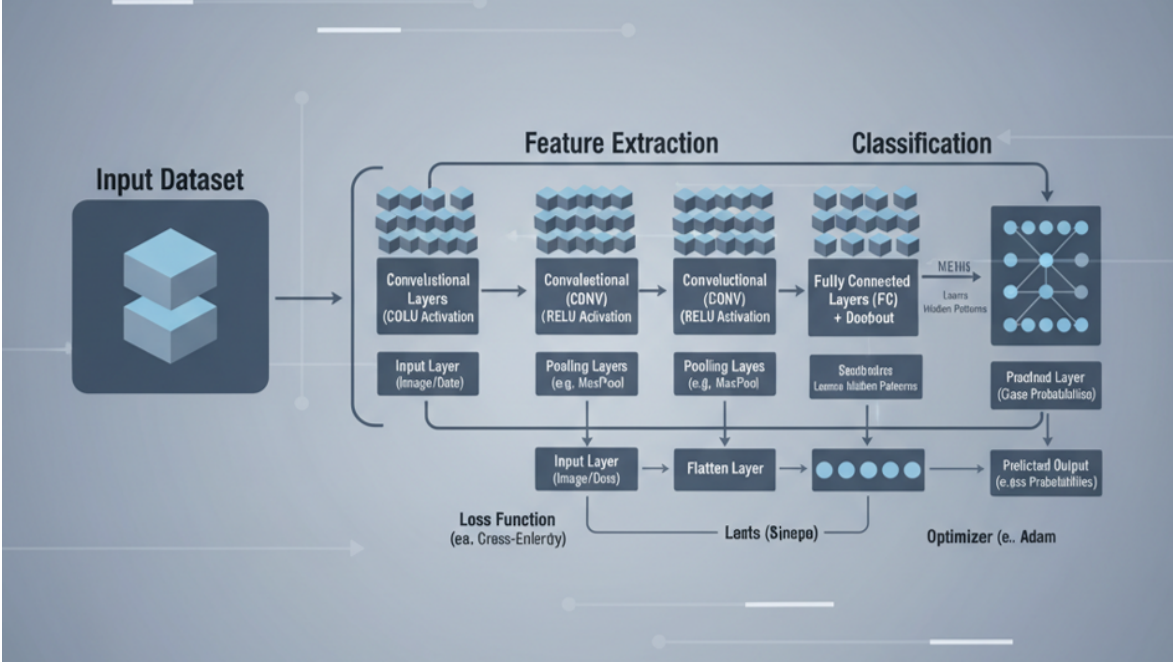


Figure 3.2: CNN Architecture

Chapter 4

System Design

4.1 Software

The software developed for this project utilizes a full-stack architecture to efficiently manage customer retention prediction workflows. At its core, the backend is powered by Flask, which orchestrates data collection, preprocessing, DNA encoding, and 2D matrix transformation before passing structured inputs to a Convolutional Neural Network (CNN) built on TensorFlow/Keras. This backend acts as the computational engine, efficiently serving predictive inference requests via RESTful API endpoints. Complementing and retrieval of raw data, encoded features, and prediction outcomes, ensuring reliability and scalability for practical deployments.

On the user-facing side, a modern React.js frontend provides interactive dashboards and detailed visualizations, enabling business users to monitor prediction results, review DNA-driven data patterns, and take data-driven retention actions. The entire platform is containerized using Docker, which streamlines deployment and maintenance across diverse environments and supports parallel scaling as analytics demand grows. With integrated real-time analytics and seamless communication between backend and frontend components, the software empowers timely, actionable insights, making advanced retention modeling accessible for subscription-based businesses.

4.1.1 Block Diagram

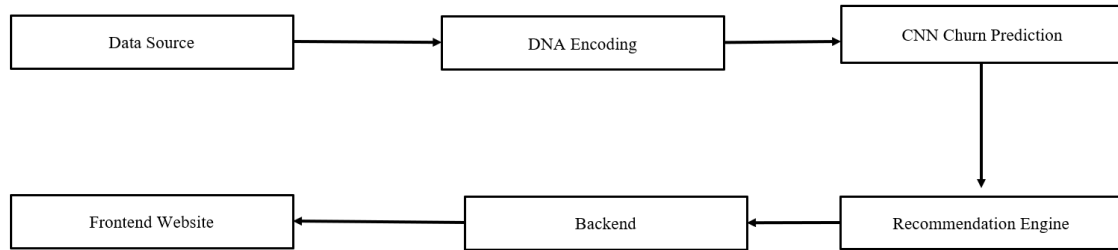


Figure 4.1: *Block diagram*

4.1.2 Working of the project

The project is organized into a modular architecture that integrates frontend, backend, and CNN to facilitate efficient customer retention prediction. In the frontend, users interact with a dashboard that presents actionable retention suggestions and visualizes DNA-encoded patterns derived from customer data. The dashboard communicates with the backend through RESTful API endpoints, enabling requests for predictions and retrieval of model outputs.

At the core of the backend lies the data preparation pipeline, which consists of three sequential steps: customer data is first collected and preprocessed, then encoded into specialized DNA-like sequences, and finally transformed into structured 2D matrices suitable for deep learning. These processed inputs are transmitted to a Convolutional Neural Network (CNN), built on TensorFlow/Keras, which learns intricate behavioral and transactional patterns to accurately predict customer churn.

All interactions, inference requests, and prediction results are handled by a Flask interface, ensuring robust, low-latency communication between the dashboard and the model. DNA feature representations, and prediction results, supporting reliable storage and quick retrieval of key data for further analysis and reporting. This comprehensive system enables seamless data flow, real-time analytics, and user-driven retention management in a scalable, deployment-ready environment.

Chapter 5

Result

The project has advanced through the foundational stages, beginning with comprehensive data collection and organization essential for subsequent processing. Initial efforts concentrated on assembling and standardizing the data source, enabling effective DNA encoding and seamless integration into the system pipeline. The core system architecture—comprising DNA encoding, CNN-based churn prediction, storage, and modular backend connectivity—has been clearly defined and partially implemented. With this groundwork established, the project is well-positioned for the next phases of model training, deployment of the recommendation engine, and frontend integration. These initial accomplishments provide a robust basis for future improvements, ensuring the system can support scalable, accurate, and user-friendly solutions for customer churn prediction and recommendation.

The modular architecture designed for this system ensures adaptability for future extensions and upgrades. By isolating critical steps—such as DNA encoding, CNN prediction, and recommendation logic—the workflow remains flexible and highly maintainable. This approach allows for integration with diverse data sources and facilitates rapid iteration in model development, supporting experimental enhancements and partially automating data-driven customization for end users. Furthermore, steady progress on backend connectivity enables reliable storage, quick retrieval, and better scalability, laying the groundwork for robust production deployment. While initial objectives have been achieved, several technical challenges and research opportunities remain. Future work will focus on optimizing the encoding and prediction pipeline for greater accuracy and efficiency, investigating advanced

recommendation strategies, and enhancing front-end usability. Collaboration with experts in machine learning, bioinformatics, and user experience design will be prioritized to refine the system. The results so far demonstrate a strong proof of concept, encouraging further exploration in real-world scenarios for improved customer engagement and retention.

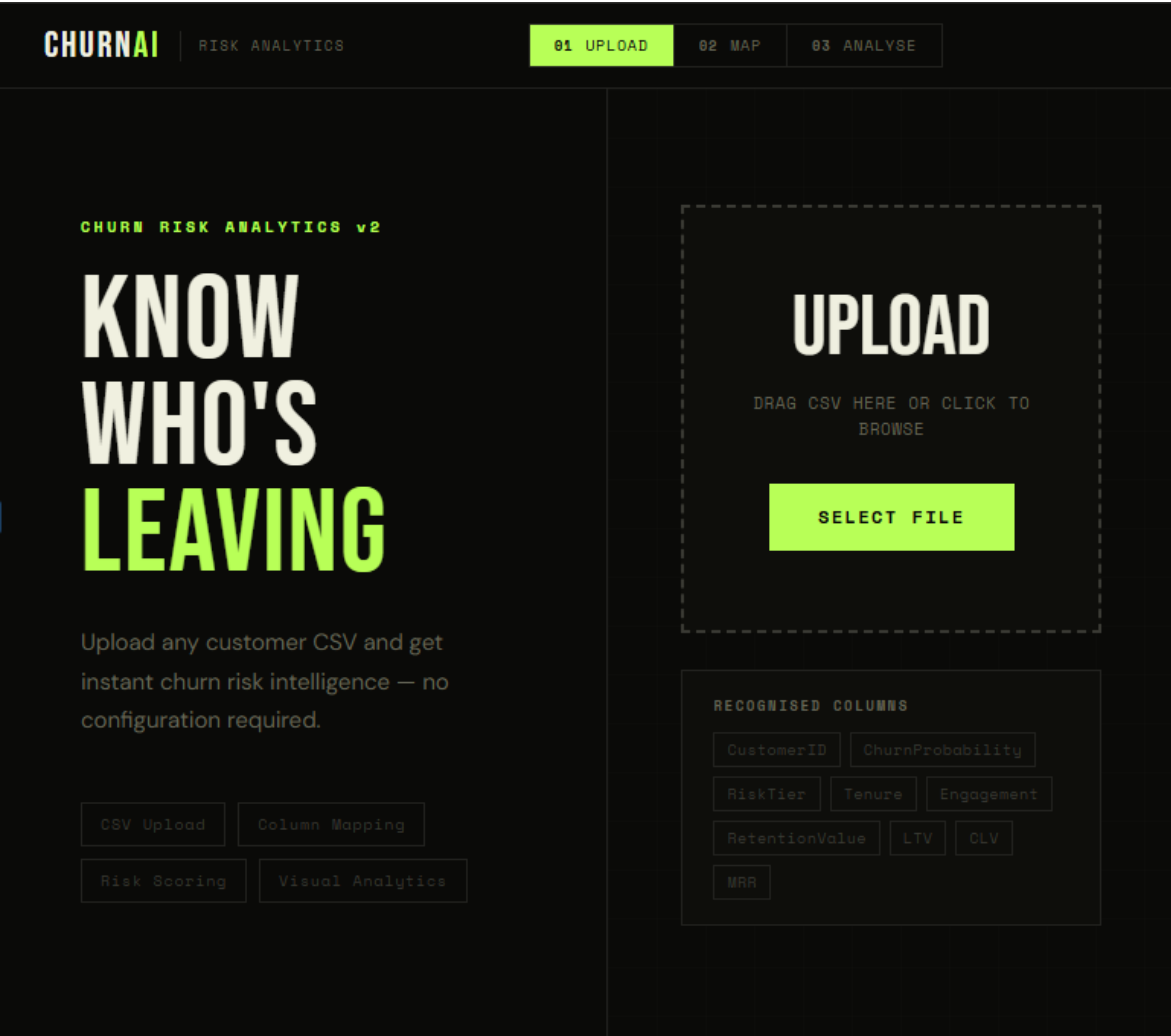


Figure 5.1: Frontend First Page

COMMUNO

RISK ANALYTICS

✓ UPLOAD

02 MAP

03 ANALYSE

STEP 02 / 03

MAP COLUMNS

churn_report_export.csv - 7 cols detected

4
COLS MAPPED

ID ✓ DONE

CUSTOMER ID

Row identifier

☐ nothing

☒ CustomerID

RISK ✓ DONE

CHURN RISK

Probability or risk tier
Numeric cols averaged; categorical auto-mapped.

☐ nothing

☒ ChurnProbability

☒ RiskTier

☐ ChurnReasons

TER ✓ DONE

TENURE

Time active with you

☒ nothing

ENG ✓ DONE

ENGAGEMENT

Activity level

☒ nothing

VAL ✓ DONE

RETENTION VALUE

Monetary value if retained

☐ nothing

☒ ExpectedRetentionGain

✓ READY TO ANALYSE

RUN ANALYSIS →

Figure 5.2: Map columns

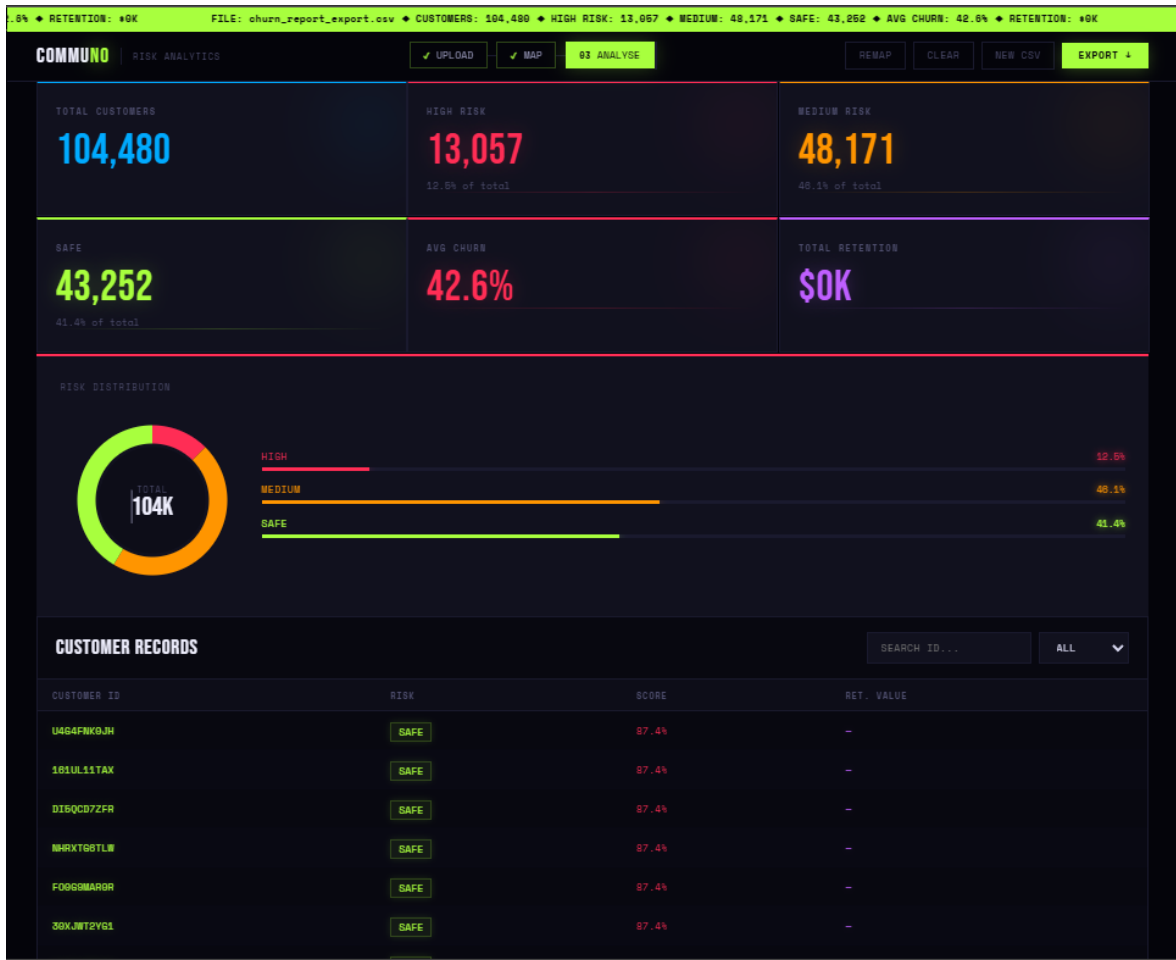


Figure 5.3: Result Frontend

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